

Joseph Thuha Maina

Geospatial Software Engineer (WebGIS & Spatial Systems)

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PROFILE

Geospatial Software Engineer with 4+ years of experience designing, building, and operating production-grade GIS platforms and large-scale spatial data pipelines. Strong focus on backend systems, GIS architectures, and cloud-deployed geospatial services. Proven track record delivering scalable, reliable GIS platforms used across 90+ country offices and supporting thousands of raster and vector datasets. Comfortable working end-to-end across data ingestion, processing, APIs, cloud infrastructure, and web-based geospatial visualization in engineering-driven teams.

CORE COMPETENCIES

- **Geospatial Software Engineering (Backend-Focused)** Design and implementation of production-grade GIS backends, spatial APIs, and service-oriented architectures.
- **Spatial Data Pipelines & ETL** Large-scale raster and vector ingestion, validation, transformation, and publishing workflows using Python and PostGIS.
- **Cloud-Deployed GIS Platforms** Design, deployment, and operation of cloud-native geospatial systems on Azure (primary), with experience across AWS and GCP.
- **Spatial Databases & Performance Optimization** Advanced PostGIS schema design, spatial indexing strategies, query optimization, and analytics-ready data modeling.
- **Web-GIS Architectures & OGC Standards** End-to-end WebGIS systems using GeoServer/GeoNode, OGC services (WMS, WFS, WMTS, XYZ), and interoperable data services.
- **Earth Observation & Large Raster Systems** Handling and visualization of EO datasets (Sentinel/Copernicus, night-time lights), dynamic raster tiling (TiTiler), and cloud-optimized formats (COG, PMTiles).
- **API Design & Distributed Systems** RESTful API development using FastAPI and Django, including asynchronous and message-based ingestion workflows.
- **DevOps & Containerization** Dockerized deployments, CI/CD pipelines, Kubernetes-based service operation, and object-storage-centric architectures.
- **Technical Leadership & Collaboration** Architecture definition, mentoring developers and analysts, documentation, and collaboration in globally distributed, remote-first teams.

PROFESSIONAL EXPERIENCE

United Nations Development Programme (UNDP), *Geospatial Developer* Jan 2022 – Dec 2025 | Remote / Multi-Country

- Core developer of GeoHub, an open-source, cloud-based geospatial platform used by 90+ UN country offices to deliver SDG analytics and operational decision-making at global scale.
- Designed and operated scalable spatial ETL pipelines using Python and PostGIS, processing 4,000+ raster and vector datasets across global humanitarian and development programs.
- Led the redesign of a critical spatial processing pipeline (Data Futures Exchange), improving processing performance by approximately 60% and significantly increasing system reliability.
- Built and maintained backend GIS APIs using FastAPI and Django, integrating spatial databases, raster services, and cloud-native components.
- Implemented dynamic raster tiling services using TiTiler, enabling efficient visualization and access to large Earth-observation datasets.
- Operated cloud-based geospatial data pipelines on Microsoft Azure, leveraging Blob Storage and Spark-based utilities for large-scale spatial processing.
- Implemented asynchronous, message-based ingestion workflows to support reliable data validation, transformation, and publishing.
- Optimized PostGIS schemas, spatial indexes, and query patterns, improving performance and stability of production services.
- Delivered technical training and onboarding to approximately 500 colleagues, enabling adoption and effective use of GeoHub across distributed country teams.
- Collaborated with globally distributed engineering and analytics teams, authored technical documentation, and supported CI/CD-driven deployments.

GoalSpatial Ltd, *Geospatial Systems Lead* Jan 2022 – Jul 2024 | Nairobi / Remote

- Led end-to-end development of production Web-GIS systems, spanning backend APIs, PostGIS databases, GeoNode/GeoServer services, and interactive web mapping interfaces.
- Designed and implemented GeoNode-based GIS platforms, including layer publishing, metadata management, access control, and exposure of GeoServer WMS/WFS/WMTS/XYZ services.
- Architected and optimized PostGIS-backed spatial data pipelines, including schema design, spatial indexing strategies, and query performance tuning.
- Built backend services using Python and Django to support spatial analysis, data ingestion, and reporting workflows.
- Developed interactive Web-GIS applications using MapLibre and Leaflet, supporting visualization of large raster and vector datasets.
- Deployed and operated GIS systems using Docker and cloud infrastructure, ensuring scalability, security, and operational reliability.
- Defined technical standards, system architecture, and development workflows; mentored developers and GIS analysts through code reviews and architectural guidance.
- Worked directly with clients to translate operational and domain requirements into robust, production-ready geospatial systems.

Kenya Red Cross Society, *GIS & Spatial Data Specialist – Disaster Response (Volunteer)* Jan 2018 – Dec 2021 | Nairobi

- Supported disaster response operations through GIS mapping and spatial data digitization under time-sensitive conditions.
- Produced geospatial datasets and maps used for field coordination and response planning.
- Worked with satellite imagery and base datasets to digitize flood-, fire-, and emergency-affected areas.
- Collaborated with humanitarian teams to deliver clear, actionable geospatial outputs in crisis contexts.

EDUCATION

Bachelor of Engineering in Geospatial Engineering, *The Technical University of Kenya*

Nairobi, Kenya

Focus areas:

- Geographic Information Systems (GIS)
- Remote Sensing
- Spatial Databases
- Cartography
- Geospatial Programming

Thesis: *A Mobile Crowdsourcing GIS for Reporting Emergent Illegal Waste Disposal Sites*

- Designed a mobile-enabled platform for capturing georeferenced field reports with structured forms and GPS validation.
- Integrated collected data into a centralized database for analysis and visualization.

TECHNICAL SKILLS

Backend & Spatial Systems

- Python (FastAPI, Django, Flask)
- REST API design and service-to-service integration
- PostgreSQL / PostGIS (schema design, spatial queries, performance tuning)
- Spatial ETL pipelines for raster and vector data (ingestion, validation, transformation, publishing)
- Analytics-ready and project-oriented geospatial data models

Geospatial & Earth Observation

- GDAL, Rasterio, GeoPandas, NumPy, Xarray, Zarr
- QGIS/ArcGIS
- Sentinel / Copernicus datasets, night-time lights
- Large-scale raster and vector data handling
- Data quality assurance, validation rules, and metadata standards

Web-GIS Platforms & Standards

- GeoServer (via GeoNode): WMS, WFS, WMTS, XYZ/TMS
- GeoNode (layer publishing, metadata, permissions)
- OGC standards and interoperable GIS service workflows
- TiTiler for dynamic raster tiling, algorithms and visualization
- Cloud optimised standards (COG, PMTiles, MBTiles)

Web Mapping & Visualization

- MapLibre, Mapbox, Leaflet, OpenLayers
- Modern front-end frameworks: React, Vue, SvelteKit
- Interactive geospatial dashboards and cartographic products

Cloud, DevOps & Infrastructure

- Docker and containerized deployments
- CI/CD pipelines (GitHub Actions)
- Cloud platforms: Microsoft Azure (primary), AWS, GCP
- Object-storage-based geospatial architectures
- Kubernetes (deployment and operation of containerized services)

LANGUAGES

English - Fluent, Swahili - Fluent, Spanish - Working Proficiency

REFERENCES

Available upon request